

Quantum Differential Equation Solver via Hybrid Oscillator-Qubit Linear Combination of Hamiltonian Simulations

Elin Ranjan Das, Muqing Zheng, Rishab Dutta, Ang Li, Yuan Liu

August 2025

Abstract

We propose an advanced method for quantum simulation based on the linear combination of Hamiltonian simulation (LCHS) within a hybrid oscillator-qubit architecture. This approach capitalizes on the inherent advantages of continuous (oscillator) and discrete (qubit) variables, eliminating the need for discretization and the associated truncation errors commonly found in discrete variable LCHS. This work marks an important step forward in overcoming the limitations of traditional simulation methods and advancing the capabilities of quantum computing.

1 Introduction

The ability of controllable quantum systems to efficiently emulate complex quantum dynamics has motivated substantial theoretical and experimental efforts. Accurate simulation of quantum many-body systems underpins advances in condensed matter physics, quantum chemistry, high-energy physics, and materials science [1]. However, realizing scalable and precise quantum simulation remains a central challenge due to hardware limitations and algorithmic constraints.

One of the most powerful algorithmic paradigms for digital quantum simulation is Hamiltonian simulation via linear combinations of unitaries (LCU) [2]. In this framework, the target Hamiltonian is decomposed into a weighted sum of efficiently implementable unitary operators, enabling high-precision time evolution with provable complexity bounds. Building on this idea, linear combination of Hamiltonian simulation (LCHS) [3, 4] methods provide a systematic way to construct evolution operators while maintaining controllable error scaling. However, existing implementations of LCHS rely exclusively on discrete-variable (qubit-based) architectures, where continuous degrees of freedom must first be discretized. This discretization introduces truncation errors and often requires large qubit overhead to achieve high fidelity, particularly when simulating systems with inherently continuous variables.

Continuous-variable (CV) quantum systems such as harmonic oscillators realized in photonic modes or superconducting cavities naturally encode infinite-dimensional Hilbert spaces. These systems are well suited for representing bosonic modes and field-theoretic models without the need for coarse discretization. Meanwhile, qubit-based platforms provide strong nonlinearity, robust control, and efficient implementation of conditional operations. Hybrid oscillator-qubit architectures combine the advantages of both paradigms, offering expanded computational expressiveness and improved resource efficiency.

In this work, we propose an advanced method for quantum simulation based on linear combination of Hamiltonian simulation within a hybrid oscillator-qubit architecture. By leveraging continuous-variable encodings for naturally bosonic degrees of freedom and discrete-variable control for selective unitary operations, our approach eliminates the discretization step that typically introduces truncation errors in purely qubit-based LCHS implementations. The resulting framework enables more faithful representation of continuous dynamics while preserving the algorithmic strengths of linear-combination-based simulation techniques.

2 Motivation

LCHS provides an optimal state preparation cost and a near-optimal scaling in matrix queries on all parameters for solving linear ordinary differential equations (ODEs) in the form of

$$\frac{du(t)}{dt} = -A(t)u(t) + b(t), \quad u(0) = u_0 \quad (1)$$

where $t \in [0, T]$, $A(t) \in \mathbb{C}^{N \times N}$ and $u(t), b(t) \in \mathbb{C}^N$. The solution of the ODE (1) has the explicit form

$$u(T) = \mathcal{T}e^{-\int_0^T A(s)ds}u_0 + \int_0^T \mathcal{T}e^{-\int_s^T A(s')ds'}b(s)ds \quad (2)$$

where $\mathcal{T}$ is the time-ordering operator. To start with, the coefficient matrix $A(t)$ is first decomposed into

$$A(t) = L(t) + iH(t)$$

via Cartesian decomposition

$$L(t) = \frac{A(t) + A^\dagger(t)}{2}, H(t) = \frac{A(t) - A^\dagger(t)}{2i}.$$

It is assumed that $L(t) \succeq 0$. Otherwise, we can shift the eigen spectrum of $u(t)$ by substituting $u(t) = e^{ct}v(t)$ and use $v(t)$ for the ODE system so that $L(t) + cI \succeq 0$, where $-c$ is the minimum of the smallest eigenvalues of $L(t)$. LCHS provides that

$$\mathcal{T}e^{-\int_0^t A(s)ds} = \int_{\mathbb{R}} g(k) \left[\mathcal{T}e^{-i\int_0^t (kL(s) + H(s))ds} \right] dk \quad (3)$$

$$\int_0^t \mathcal{T}e^{-\int_s^t A(s')ds'}b(s)ds = \int_0^t \int_{\mathbb{R}} g(k) \left[\mathcal{T}e^{-i\int_s^t (kL(s') + H(s'))ds'} \right] b(s)dkds \quad (4)$$

where $g(k) := \frac{f(k)}{1-ik}$ for some kernel function $f(k)$ and $t \in [0, T]$. The choice of $f(k)$ that leads to near-optimal complexity dependency on all parameters is $f(k) = \frac{e^{2\beta}}{2\pi e^{(1+ik)^\beta}}$, where $\beta \in (0, 1)$ is a user-defined constant [4]. Note that the terms in brackets in Eq. (3) and (4) are unitary. Thus, LCHS further truncates and discretizes the integrals in Eq. (3) and (4) into a sum of unitaries, employing state preparation circuits for $|u_0\rangle := \frac{u_0}{\|u_0\|}$ and $|b(s)\rangle := \frac{b(s)}{\|b(s)\|}$ and the linear combination of unitaries (LCU) circuits to prepare the solution state $|u(T)\rangle := \frac{u(T)}{\|u(T)\|}$ [3, 4].

Combining with the finite difference method and linearization methods, for example, Carleman linearization [5], various non-linear partial differential equations can also be converted into a linear ODE system that is solvable by LCHS. However, the number of terms in the discretized LCHS integral, denoted by M , brings a huge cost on quantum resources in the circuit implementation level for either near-term or early fault-tolerant quantum machines. For the complementary solution, M has complexity

$$M \in \mathcal{O} \left(T \max_t \|L(t)\| \left(\log \frac{1}{\epsilon} \right)^{1+1/\beta} \right)$$

where ϵ is the total error incurred by LCHS [4]. The corresponding LCU circuit requires $\lceil \log_2(M) \rceil$ ancillary qubits, and more problematically, $\lceil \log_2(M) \rceil$ -controlled multiple-qubit unitary operators for Hamiltonian simulations. In this case, for a time-independent ODE, the value of M can be in the 10^6 magnitude when ϵ is around 10^{-5} , $\beta = 0.8$, $T = 1000$, and $\|L\| = 1$. More exact scalings of LCHS are provided in [6].

Inspired by the continuous LCU in [7], a hybrid CV-DV LCHS is a promising direction to bring down the resource cost and, ideally, remove truncation and discretization errors in the integrals calculation. Specifically, continuous LCU can use $\mathcal{O}(1)$ number of ancillary oscillators, along with CV state preparation and CV measurements, replace $\lceil \log_2(M) \rceil$ ancillary qubits in the DV LCU circuit, while the unitaries for Hamiltonian evolution remain in qubit space. Continuous LCU circuit can compute the integrals in Eq. (3) and (4) without truncation and discretization on integrals, whereas truncations errors may still persist in CV state preparation and hybrid Hamiltonian simulation.

3 Methodology

Let us consider a time-independent homogeneous ODE system, where $A(t) := A$ and $b(t) := 0$ for all t . From Eq. (2) and (3), the complementary solution becomes

$$u(t) = \left[\int_{\mathbb{R}} g(k) e^{-it(kL+H)} dk \right] u_0 \quad (5)$$

for $t \in [0, T]$. Using a single oscillator for the ancillary in continuous LCU, the qubits that synthesize unitary $e^{-it(kL+H)}$ need to couple with this ancillary oscillator, leading to the joint evolution $e^{-it(L \otimes \hat{k} + H \otimes I)}$, where $\hat{k}$ is

one of the canonical conjugation operators ($\hat{x}$ and $\hat{p}$) and I is the identity operator. That is to say, the qumode remains untouched when H acts on the qubit system.

Before forming the circuit for the joint evolution, an appropriate function $g(k)$ has to be obtained. To do this, we need to prepare and post-select two appropriate CV states.

Let $|\Psi\rangle = |\psi\rangle_{\text{osc}} |u_0\rangle_q$ be the preparation state, where $|\psi\rangle_{\text{osc}}$ is the initial oscillator state and $|u_0\rangle$ is the initial qubit state. Let $\langle\Phi| = \langle\phi|_{\text{osc}} \otimes \mathbf{I}_q$ be the post-selection projector, where $\langle\phi|_{\text{osc}}$ acts only on the oscillator and the qubit remains untouched.

If we choose the position operator $\hat{x}$ for k , then according to ref. [7], the qumode sandwich gives

$$\langle\Phi| e^{-it(\hat{x} \otimes L + \mathbf{I}_{\text{osc}} \otimes H)} |\Psi\rangle = \left[\int_{\mathbb{R}} dx \phi^*(x) \psi(x) e^{-it(xL+H)} \right] |u_0\rangle_q. \quad (6)$$

Comparing to the Eq. (5), this gives $g(x) = \phi^*(x)\psi(x)$. Namely, we need to engineer the preparation and post-selection states to obtain an appropriate $g(x)$ that satisfies LCHS's requirements.

We choose the post-selection state $|\phi\rangle$ to be a squeezed vacuum, whose position-space wavefunction is

$$\phi_r(x) := S(r, 0)\phi_0(x) = \left(\frac{2}{e^{2r}\pi}\right)^{1/4} e^{-x^2/e^{2r}} = \left(\frac{2}{\sigma^2\pi}\right)^{1/4} e^{-x^2/\sigma^2} \quad (7)$$

with squeezing parameter r and wavefunction standard deviation $\sigma = e^r$, and this gives the corresponding initial state to be prepared

$$\psi_r(x) := \left(\frac{\sigma^2\pi}{2}\right)^{1/4} g(x) e^{x^2/\sigma^2} = \mathcal{N} g(x) e^{x^2/\sigma^2}. \quad (8)$$

This leads to our formulation of a CV/DV LCHS theorem:

Theorem 1 (Continuous-discrete variable LCHS). *Suppose $A = L + iH$ and $L \succeq 0$, then $u(t)$ can be expressed as the following:*

$$|u(t)\rangle = e^{-At} |u(0)\rangle = \langle\Phi| e^{-it(\hat{x} \otimes L + \mathbf{I}_{\text{osc}} \otimes H)} |\Psi\rangle \quad (9)$$

$$= (\langle\phi|_{\text{osc}} \otimes \mathbf{I}_q) e^{-it(\hat{x} \otimes L + \mathbf{I}_{\text{osc}} \otimes H)} (|\psi\rangle_{\text{osc}} \otimes |u_0\rangle_q) \quad (10)$$

$$= \left[\int_{\mathbb{R}} dx \phi^*(x) \psi(x) e^{-it(xL+H)} \right] |u_0\rangle_q. \quad (11)$$

where $|\psi\rangle_{\text{osc}}$ and $|\phi\rangle_{\text{osc}}$ are squeezed Fock states.

At this point, the missing piece for the whole hybrid CV-DV LCHS circuit is the oracle syntheses for matrices L and H , whose basis-gate implementations are highly dependent on the structures of both matrices and are very problem-specific. Many previous works in the DV regime tackle this problem by decomposing coefficient matrices into linear combinations of Pauli matrices, for example, hyperbolic partial differential equations [8] and advection-diffusion equation [9]. Borrowing their designs and following the circuit synthesis in [7], the evolution of decomposed Hamiltonians coupled with oscillators can be constructed in CV-DV circuits.

3.1 State Preparation

We now consider two special cases of $\psi_r(x)$ preparation and their implications.

- $\mathbf{r} = \mathbf{0}$ ($\sigma = 1$): This leads to the post-selection state being the vacuum state $\phi_0(x)$. However, our preparation state becomes

$$\psi_0(x) := \mathcal{N} g(x) e^{x^2} \quad (12)$$

which is not a physically realizable state, as $\psi_0(x) \notin L^2(\mathbb{R})$.

This gives us an additional condition imposed on the $g(x)$ for CV-DV LCHS:

$$\exists r > 0 \text{ s.t. } \int_{\mathbb{R}} dx |g(x)|^2 e^{\frac{2x^2}{\sigma^2}} < \infty \quad (13)$$

- $\mathbf{r} \rightarrow \infty$ ($\sigma \rightarrow \infty$): This leads to $\phi_r(x)$ being the infinitely squeezed vacuum state. Then our preparation state is simply the kernel itself:

$$\psi_\infty(x) := \lim_{r \rightarrow \infty} \psi_r(x) = \mathcal{N} g(x) \quad (14)$$

The initial state $|\psi_r\rangle$ can be approximated via a finite squeezed Fock expansion

$$\psi_r(x) = \mathcal{N}g(x)e^{x^2/\sigma^2} \approx \sum_{\mathbf{n}=0}^N C_{\mathbf{n}} \langle x|S(r, 0)|\mathbf{n}\rangle = \psi'(x) \quad (15)$$

where coefficients $\{C_{\mathbf{n}}\}$ are overlaps

$$C_{\mathbf{n}} = \langle \mathbf{n}|S^\dagger(r', 0)|\psi_r\rangle = \int_{\mathbb{R}} dx \phi_{\mathbf{n}}^*(x/\sigma') e^{-x^2/\sigma'^2} \psi_r(x) \quad (16)$$

This expression simplifies to

$$C_{\mathbf{n}} = \sqrt{\frac{\sigma}{\sigma'}} \frac{1}{\sqrt{2^{\mathbf{n}} \mathbf{n}!}} \int_{\mathbb{R}} dx H_{\mathbf{n}}(x/\sigma') g(x) e^{-\gamma x^2} = \int_{\mathbb{R}} \frac{dx e^{2\beta} H_{\mathbf{n}}(x/\sigma') e^{-\gamma x^2}}{2\pi e^{(1+ix)^\beta} (1-ix)} \quad (17)$$

where $\gamma = \frac{1}{\sigma'^2} - \frac{1}{\sigma^2} = \frac{1}{e^{2r'}} - \frac{1}{e^{2r}}$, $r' < r$. While eq. (17) can have analytical forms depending on specific values of β , it is not possible to derive a generalized expression. Two examples of closed analytic form for the limit values of $\beta = \{0, 1\}$ are given in A.4.

3.2 Error bound estimation for state preparation

Ideally we want to realize $\psi_r(x) = g(x)e^{x^2/e^{2r}}$ as close as possible using Gaussian (Fock) states. Previously, authors from [10] showed that sub-optimal kernel function $f(k) = \frac{1}{\pi(1+ik)}$ could be realized with a squeezed vacuum state with maximum fidelity of 98.6% with a squeezing parameter of 0.925. However, we show that the near-optimal kernel function can be realized instead with a superposition of squeezed Fock states for arbitrary closeness, where the approximation error is bounded by the Fock cutoff level (or the oscillator energy) and the squeezing parameter:

Theorem 2. *Let $\psi_r(k)$ denote the ideal prepared state, and let $\psi'(k)$ be an ϵ_s -approximation to $\psi_r(k)$. Then for any $\epsilon_s > 0$, there exist a Fock cutoff $N \in \mathbb{N}$ and a squeezing parameter $r > 0$ such that*

$$\|\psi' - \psi_r\| < \epsilon_s.$$

Moreover, the required resources scale asymptotically as

$$N = \mathcal{O}\left(\log^2 \frac{1}{\epsilon_s}\right), \quad r = \Omega\left(\log \frac{1}{\epsilon_s}\right).$$

Proof. Let $\mathbb{H}_N = \text{span}\{\psi_{\mathbf{n}}\}_{\mathbf{n}=0}^N$ denote the truncated bosonic Hilbert space spanned by the squeezed Fock basis $\psi_{\mathbf{n}}$, and let $\Pi_N^F : L^2(\mathbb{R}) \rightarrow \mathbb{H}_N$ be the orthogonal projection. Define the truncated approximation as the projection of g ,

$$\Pi_N^F g = \sum_{\mathbf{n}=0}^N C_{\mathbf{n}} \psi_{\mathbf{n}} = \psi'.$$

The ideal prepared state is $\psi_r(k) = g(k)e^{k^2 e^{-2r}}$.

Using the triangle inequality,

$$\|\psi' - \psi_r\| \leq \|g - \Pi_N^F g\| + \|g - g e^{k^2 e^{-2r}}\| =: \tau(N) + \delta(r).$$

We estimate the two contributions separately.

g is analytic in the strip $\mathcal{S}_\rho := \{z \in \mathbb{C} : \Im(z) \in [-\rho, \rho]\}$ for some $\rho = 1 - \epsilon$ with $\epsilon \rightarrow 0$. Then the convergence analysis of Hermite approximations [11, Thm. 3.8]¹ implies that there exists a constant $\mathcal{K} > 0$ such that

$$\tau(N) = \|g - \Pi_N^F g\| \leq \mathcal{K} e^{-\rho\sqrt{2N}}.$$

Using the Taylor expansion $e^{k^2 e^{-2r}} = 1 + k^2 e^{-2r} + \mathcal{O}(k^4 e^{-4r})$ as $r \rightarrow \infty$, we obtain

$$\delta(r) = \|g(k) - g(k)e^{k^2 e^{-2r}}\| = \|-g(k)k^2 e^{-2r}\| \leq \mathcal{M} e^{-2r}$$

¹The strict version of the theorem states that g must satisfy some growth conditions as well. However, the authors imply in the same work that the growth conditions can be relaxed [11, pp. 15]. Here we assume the relaxed version of the theorem.

for some constant $\mathcal{M} > 0$ independent of r . Combining the two bounds gives $\|\psi' - \psi_r\| \leq \mathcal{M}e^{-2r} + \mathcal{K}e^{-\rho\sqrt{2N}}$. To ensure $\|\psi' - \psi_r\| < \epsilon_s$, it suffices to choose r and N such that

$$\mathcal{M}e^{-2r} \leq \epsilon_s/2, \quad \mathcal{K}e^{-\rho\sqrt{2N}} \leq \epsilon_s/2.$$

The first condition implies

$$r \geq (1/2) \log(2\mathcal{M}/\epsilon_s), \quad r \geq (1/2) \log(2\mathcal{M}/\epsilon_s),$$

hence $r = \Omega(\log(1/\epsilon_s))$.

The second condition yields $\sqrt{2N} \geq (1/\rho) \log(2\mathcal{K}/\epsilon_s)$, which implies $N \geq (1/(2\rho^2)) \log^2(2\mathcal{K}/\epsilon_s)$. Therefore $N = \mathcal{O}(\log^2(1/\epsilon_s))$. This establishes the claimed resource scaling. $\square$

The CV state preparation is well-known by the Law-Eberly Protocol (LEP) [12, 13]. The experimental implementation is thoroughly discussed in 4.1. An illustrative circuit of the hybrid LCHS is demonstrated in fig. 1. After we prepare the initial state, we move on to the Hamiltonian evolution.

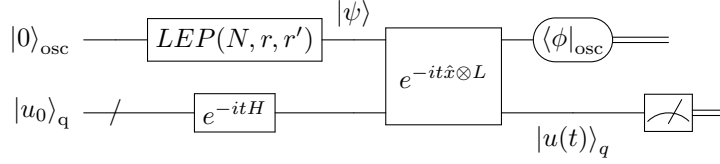


Figure 1: Workflow circuit for CVDV LCHS

3.3 Trotter-Suzuki approximation for Hamiltonian evolution

Since the hybrid Hamiltonian $\hat{x} \otimes \hat{L} + I \otimes \hat{H}$ is not directly realizable as a native interaction, its time evolution is synthesized via a Trotter-Suzuki decomposition into a sequence of implementable operations.

Let $\delta t = t/n$, where t is the total simulation time and n is the number of Trotter steps, and define

$$A = \hat{x}_N \otimes \hat{L}, \quad B = I \otimes \hat{H}.$$

where the operator $\hat{x}_N$ is understood on a suitably truncated or energy-constrained CV subspace. The first-order Lie-Trotter formula yields $U(t) = e^{-it(A+B)} \approx (e^{-i\delta t A} e^{-i\delta t B})^n$, with error scaling as $\epsilon_t = \mathcal{O}\left(\frac{t^2}{n} \|[A, B]\|\right)$.

Since $[A, B] = \hat{x}_N \otimes [\hat{L}, \hat{H}]$, we obtain the bound $\|[A, B]\| \leq \|\hat{x}_N\| \|[\hat{L}, \hat{H}]\|$. Consequently,

$$n = \mathcal{O}\left(\frac{t^2}{\epsilon_t} \|\hat{x}_N\| \|[\hat{L}, \hat{H}]\|\right). \quad (18)$$

Suppose that the Hamiltonians admit Pauli decompositions of the form

$$\hat{L} = \sum_{i=1}^{N_L} \alpha_i P_i, \quad \hat{H} = \sum_{j=1}^{N_H} \beta_j P'_j, \quad (19)$$

where P_i and P'_j are Pauli strings. Each exponential can then be further decomposed using a first-order product formula as

$$e^{-i\delta t A} \approx \prod_{i=1}^{N_L} e^{-i\delta t \alpha_i (\hat{x}_N \otimes P_i)}, \quad e^{-i\delta t B} \approx \prod_{j=1}^{N_H} e^{-i\delta t \beta_j (I \otimes P'_j)}. \quad (20)$$

Combining these decompositions yields the approximation

$$U(t) \approx \left(\prod_{i=1}^{N_L} e^{-i\delta t \alpha_i (\hat{x}_N \otimes P_i)} \prod_{j=1}^{N_H} e^{-i\delta t \beta_j (I \otimes P'_j)} \right)^n, \quad (21)$$

up to additional error arising from the non-commutativity of the Pauli terms.

Each Pauli rotation $e^{-i\theta P}$ is implemented using a standard basis-change and CNOT-ladder construction, requiring $2(w(P) - 1)$ CNOT gates, where $w(P)$ denotes the Pauli weight. Consequently, the total number of CNOT gates scales as

$$N_{\text{CNOT}} = \left(2n \sum_{k=1}^{N_L + N_H} (w(P_k) - 1) \right), \quad (22)$$

where $\{P_k\}$ denotes the collection of Pauli strings appearing in the decomposition of $\hat{L}$ and $\hat{H}$.

In addition, each Trotter step contains N_L hybrid CV-DV interactions of the form $e^{-i\delta t \alpha_i (\hat{x}_N \otimes P_i)}$, yielding a total of $N_{\text{hyb}} = nN_L$ hybrid gates.

Defining the Pauli 1-norms

$$\|\hat{L}\|_1 := \sum_i |\alpha_i|, \quad \|\hat{H}\|_1 := \sum_j |\beta_j|, \quad (23)$$

and using the bound $\|[\hat{L}, \hat{H}]\| \leq 2\|\hat{L}\|_1 \|\hat{H}\|_1$, the first-order Trotter error implies

$$n = O\left(\frac{t^2}{\epsilon_t} \|\hat{x}_N\| \|\hat{L}\|_1 \|\hat{H}\|_1\right). \quad (24)$$

Therefore, the overall CNOT complexity scales as

$$N_{\text{CNOT}} = O\left(\frac{t^2}{\epsilon_t} \|\hat{x}_N\| \|\hat{L}\|_1 \|\hat{H}\|_1 \sum_k w(P_k)\right). \quad (25)$$

Higher-order product-formula simulation. For an even order $p = 2k$, the p -th order Suzuki formula yields

$$U(t) = e^{-it(\hat{x}_N \otimes \hat{L} + I \otimes \hat{H})} \approx (S_p(\delta t))^n, \quad (26)$$

with error scaling

$$\epsilon_t = \mathcal{O}\left(\frac{t^{p+1}}{n^p} \|\hat{x}_N\| \Gamma_p(\hat{L}, \hat{H})\right), \quad (27)$$

For operators X and Y , we define the r -fold nested commutator recursively by

$$[X^{(0)}, Y] := Y, \quad [X^{(r)}, Y] := [X, [X^{(r-1)}, Y]]. \quad (28)$$

Then we can define $\Gamma_p(\hat{L}, \hat{H})$ in terms of nested commutators as

$$\Gamma_p(\hat{L}, \hat{H}) = \max_{\substack{r+s=p \\ r,s \geq 0}} \left\{ \|[\hat{L}^{(r)}, [\hat{H}^{(s)}, \hat{L}]]\|, \|[\hat{L}^{(r)}, [\hat{H}^{(s)}, \hat{H}]]\| \right\}. \quad (29)$$

For practical purposes, this quantity admits the upper bound

$$\Gamma_p(\hat{L}, \hat{H}) \leq C_p (\|\hat{L}\| + \|\hat{H}\|)^{p-1} \|\hat{L}, \hat{H}\|. \quad (30)$$

Consequently,

$$n = \mathcal{O}\left(t^{1+1/p} \left(\frac{\|\hat{x}_N\| \Gamma_p(\hat{L}, \hat{H})}{\epsilon_t}\right)^{1/p}\right). \quad (31)$$

Moreover,

$$[\hat{x}_N \otimes \hat{L}, I \otimes \hat{H}] = \hat{x}_N \otimes [\hat{L}, \hat{H}], \quad (32)$$

which implies the bound

$$\|[\hat{x}_N \otimes \hat{L}, I \otimes \hat{H}]\| \leq \|\hat{x}_N\| \|\hat{L}, \hat{H}\|, \quad (33)$$

where $\|\hat{x}_N\|$ is evaluated on a suitably truncated or energy-constrained CV subspace.

Gate counts. Let m_p denote the number of exponentials appearing in a single application of the p -th order Suzuki formula $S_p(\delta t)$. For the standard recursive construction, one has

$$m_2 = 3, \quad m_{2k} = 5 m_{2k-2}, \quad \Rightarrow \quad m_p = 3 \cdot 5^{(p/2)-1}. \quad (34)$$

The total number of CNOT gates scales as

$$N_{\text{CNOT}} = \left(2nm_p \sum_{k=1}^{N_L+N_H} (w(P_k) - 1)\right), \quad (35)$$

where $\{P_k\}$ denotes the collection of Pauli strings arising from the Pauli decompositions of $\hat{L}$ and $\hat{H}$.

Moreover, each exponential of the form $e^{-i\theta(\hat{x} \otimes P_i)}$ contributes a hybrid CV-DV interaction. Let c_p denote the number of such hybrid exponentials within a single application of $S_p(\delta t)$. For the standard Suzuki construction, one has $c_p = \Theta(m_p)$, and therefore the total number of hybrid gates scales as $N_{\text{hyb}} = O(nc_p N_L)$.

In the following section, we outline the protocol for one toy example of damped harmonic oscillator and another example of d -dimensional heat conduction equation.

3.4 Simulation Method for general linear DEs by CV-DV LCHS

3.4.1 Toy Example: Damped Harmonic Oscillator

Consider a damped harmonic oscillator with position $x(t)$, damping rate $\gamma \geq 0$, and angular frequency $\omega > 0$. Its equation of motion is

$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega^2 x = 0. \quad (36)$$

Introducing the velocity variable $v = \dot{x}$, this second-order equation can be written as a first-order linear system

$$\frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = - \begin{bmatrix} \gamma/2 & -\omega \\ \omega & \gamma/2 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}. \quad (37)$$

We denote the coefficient matrix by A .

To cast the dynamics into the LCHS framework, we perform the Cartesian decomposition of A ,

$$L = \frac{A + A^\dagger}{2} = \frac{\gamma}{2} I, \quad (38)$$

$$H = \frac{A - A^\dagger}{2i} = -\omega Y, \quad (39)$$

where I is the identity matrix and Y is the Pauli- Y matrix.

The corresponding unitary operators appearing in the LCHS circuit construction are therefore

$$\exp(-iT k L) = \exp\left(-iI \frac{\gamma T k}{2}\right), \quad (40)$$

$$\exp(-iT H) = \exp(i\omega T Y). \quad (41)$$

Thus, the dissipative contribution is captured by a uniform decay factor, while the Hamiltonian part corresponds to a planar rotation generated by Y . The resulting LCHS simulation can be implemented using single-qubit R_Y rotations together with non-unitary damping steps, with standard Trotterization over discrete time steps.

3.4.2 Linear 1st order PDE Example: Discretized One-dimensional Heat Equation

Consider one-dimensional (1D) heat equation

$$\frac{\partial u(x, t)}{\partial t} = \alpha \frac{\partial^2 u(x, t)}{\partial x^2} \quad (42)$$

where α (or often written in a squared constant) is called the thermal diffusivity of a length- L rod. Its initial condition is

$$u(x, 0) = f(x), 0 < x < L.$$

And we start with the easiest boundary condition, a homogeneous Dirichlet boundary condition

$$u(0, t) = u(L, t) = 0, t > 0.$$

With a 6-point (2 end-points, x_0 and x_5 are fixed by the boundary condition) second order central difference approximation

$$\left. \frac{\partial^2 u(x, t)}{\partial x^2} \right|_{x_j} \approx \frac{u(x_{j+1}) - 2u(x_j) + u(x_{j-1}))}{h^2}$$

we have

$$\frac{d}{dt} \begin{bmatrix} u(x_1, t) \\ u(x_2, t) \\ u(x_3, t) \\ u(x_4, t) \end{bmatrix} = -\frac{\alpha}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} u(x_1, t) \\ u(x_2, t) \\ u(x_3, t) \\ u(x_4, t) \end{bmatrix}, u(x, 0) = f(x) \quad (43)$$

And to decompose the coefficient matrix into Pauli words,

$$T = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = 2I_0 I_1 - I_0 X_1 - \frac{1}{2} (X_0 X_1 + Y_0 Y_1) \quad (44)$$

And $A = \frac{\alpha}{\hbar^2} T$. Since A is real symmetric,

$$L = \frac{A + A^\dagger}{2} = A, \quad H = \frac{A - A^\dagger}{2i} = \mathbf{0} \quad (45)$$

While we have a Trotter resource estimation based on the Pauli decomposition of T , for the heat equation, we can utilize specific structural leverage of T to propose an alternative protocol and resource estimation. We outline this below.

For a grid of M data points, we have a register of $m = \lceil \log_2 M \rceil$ qubits. Then our computational basis on m qubits is $\{|x\rangle\}_{x=0}^{2^m-1}$. Since the 1D Laplacian T is a symmetric tridiagonal Toeplitz matrix, we can decompose this as

$$T = 2 \sum_{x=0}^{2^m-1} |x\rangle\langle x| - \sum_{x=0}^{2^m-2} (|x\rangle\langle x+1| + |x+1\rangle\langle x|), \quad (46)$$

Let us define a binary increment operator or shift operator S as follows:

$$\hat{S} := \sum_{x=0}^{2^m-2} |x+1\rangle\langle x|, \quad \hat{S}^\dagger := \sum_{x=0}^{2^m-2} |x\rangle\langle x+1|$$

where

$$\hat{S} |y\rangle = |y+1\rangle, \quad \hat{S}^\dagger |y+1\rangle = |y\rangle, \quad y \in \{0, 1, \dots, 2^m-2\}.$$

Then the Laplacian can be expressed as

$$T = 2I - (\hat{S} + \hat{S}^\dagger) \quad (47)$$

For a discrete time step $\delta t = t/n$, the evolution is given by:

$$\mathcal{U}\left(\frac{t}{n}\right) \approx \exp\left(i\theta \hat{x} \otimes \hat{S}\right) \exp\left(i\theta \hat{x} \otimes \hat{S}^\dagger\right) \exp\left(-i2\theta \hat{x} \otimes I\right) \quad (48)$$

where $\theta = \frac{t\alpha}{nh^2}$.

We realize the shift operator $\hat{S}$ using multicontrolled NOT (MCX) gates, which are not native. There are numerous ways to realize the shift operator. For example: We can realize the incrementer with $3m$ Toffoli gates, $\mathcal{O}(m)$ X gates, and $\log^* m$ clean ancilla qubits [14]. While the ancilla register size theoretically seems to scale logarithmically with the system register size, the properties of $\log^*$ function restricts the practical size to be no more than 4 qubits. Then each Trotter step requires $12m$ Toffoli gates, m CD gates each on m qubits $e^{-i\frac{2t\alpha q_j}{nh^2} \hat{x} \otimes Z}$, $2m$ CD gates each on j^{th} qubit ($e^{i\frac{t\alpha 2^j q_j}{nh^2} \hat{x} \otimes Z}$), and $\mathcal{O}(m)$ circuit depth. The total resource count for p^{th} order Trotterization is given in the first row of table 1. The circuit is given in figure 3.

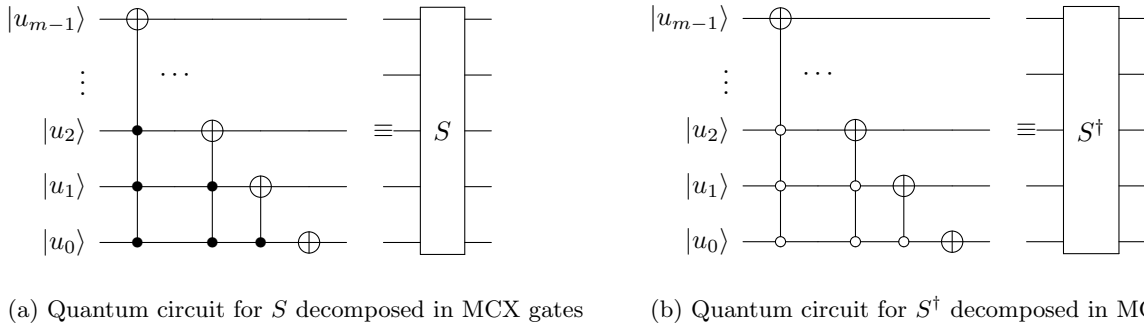


Figure 2: Building blocks of the quantum circuit

Alternatively, we can couple the system oscillator to an ancilla oscillator, in which case the ancilla oscillator encodes the spatial basis x in the position operator $\hat{x}_2$. Then the heat equation becomes

$$\frac{d\mathbf{u}}{dt} = -\alpha \hat{p}_2^2 \mathbf{u} \quad (49)$$

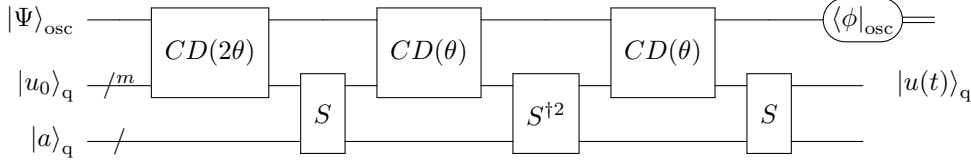


Figure 3: Quantum circuit for one complete Trotter step in 1D heat equation using ancilla register and circuit blocks from fig. 2.

where $\mathbf{u} := \int u(x, t) |x\rangle dx = |\psi\rangle_{\text{osc}_1} |u(t)\rangle_{\text{osc}_2}$. This qumode embedding scheme is already known [10]. However, we can prepare the initial state $\mathbf{u}(t=0)$ using our proposed methodology from 3.1. Then from theorem 2, we can obtain the evolution:

$$|u(t)\rangle = (\langle\phi|_{\text{osc}} \otimes \mathbb{I}) \hat{U} (|\psi\rangle_{\text{osc}} \otimes |u_0\rangle_{\text{anc}}) \quad (50)$$

where $\hat{U} := R(-\frac{\pi}{2}) e^{-i\alpha t \hat{x}_1 \otimes \hat{x}_2^2} R(\frac{\pi}{2})$ with $e^{-i\alpha t \hat{x}_1 \otimes \hat{x}_2^2}$ being the cubic-QND gate [15]. This requires two additional ancilla modes.

3.4.3 Example: Two dimensional Heat Equation

Two dimensional heat equation

$$\frac{\partial u(x, y, t)}{\partial t} = \alpha \left(\frac{\partial^2 u(x, y, t)}{\partial x^2} + \frac{\partial^2 u(x, y, t)}{\partial y^2} \right), u(x, y, 0) = f(x, y) \quad (51)$$

for $0 < x < a$ and $0 < y < b$. With homogeneous Dirichlet boundary conditions

$$u(0, y, t) = u(a, y, t) = 0 \leq y \leq b, t \geq 0 \quad (52)$$

$$u(x, 0, t) = u(x, b, t) = 0 \leq x \leq a, t \geq 0 \quad (53)$$

Again, using central difference method, we have

$$\frac{du_{i,j}(t)}{dt} \approx \frac{u_{i+1,j}(t) - 2u_{i,j}(t) + u_{i-1,j}(t)}{h_x^2} + \frac{u_{i,j+1}(t) - 2u_{i,j}(t) + u_{i,j-1}(t)}{h_y^2} \quad (54)$$

where $u_{i,j}(t) = u(x_i, y_j, t)$.

Let M be the number of interior points (boundary points are determined by boundary condition)

$$M = \lceil a/h_x - 1 \rceil \cdot \lceil b/h_y - 1 \rceil. \quad (55)$$

Suppose there 2 interior points in x -direction and 4 interior points in y -direction, we denote

$$u_1(t) = u(x_1, y_1, t), u_2(t) = u(x_1, y_2, t), \dots \quad (56)$$

$$u_5(t) = u(x_2, y_1, t), u_6(t) = u(x_2, y_2, t), \dots \quad (57)$$

Note that in this scheme, the neighbor point in y direction is ± 1 indices, but neighbor in x direction is $\pm \lceil b/h_y - 1 \rceil$ indices.

For the form of $u_t = -Au(t)$, in our ordering scheme $A_{k,k} = \frac{2}{h_x^2} + \frac{2}{h_y^2}$, the neighbor in x direction is on the super and sub diagonal $A_{k\pm 1,k} = -\frac{1}{h_x^2}$, and the neighbor in y direction is $A_{k,k\pm 4} = -\frac{1}{h_y^2}$.

An example coefficient matrix is

$$A = \begin{pmatrix} \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & 0 \\ -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 \\ 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 \\ 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & 0 & 0 & 0 & -\frac{1}{h_x^2} \\ -\frac{1}{h_x^2} & 0 & 0 & 0 & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 \\ 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 \\ 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} \\ 0 & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} \end{pmatrix}.$$

The Pauli decomposition is

$$A = \frac{1}{h_y^2} I \otimes T + \frac{2}{h_x^2} I_0 I_1 I_2 - \frac{1}{h_x^2} X_0 I_1 I_2 \quad (58)$$

$$= \frac{1}{h_y^2} \left[2I_0 I_1 I_2 - I_0 I_1 X_2 - \frac{1}{2} (I_0 X_1 X_2 + I_0 Y_1 Y_2) \right] + \frac{2}{h_x^2} I_0 I_1 I_2 - \frac{1}{h_x^2} X_0 I_1 I_2 \quad (59)$$

$$= \left(\frac{2}{h_y^2} + \frac{2}{h_x^2} \right) I_0 I_1 I_2 - \frac{1}{h_y^2} \left(I_0 I_1 X_2 - \frac{1}{2} (I_0 X_1 X_2 + I_0 Y_1 Y_2) \right) - \frac{1}{h_x^2} X_0 I_1 I_2 \quad (60)$$

We take a qubit register of m qubits and divide it into two subregisters of m_x and m_y qubits. We label them as X register and Y register. If we choose arbitrary grid resolutions on both dimensions so that we have $M_x = 2^{m_x}$ points along the x -dimension and $M_y = 2^{m_y}$ points along the y -dimension, then we can map the system to the subregisters. The 2D Laplacian operator A is then defined by:

$$A = \frac{1}{h_x^2} (T_x \otimes I^{\otimes m_y}) + \frac{1}{h_y^2} (I^{\otimes m_x} \otimes T_y)$$

where T_x and T_y are the 1D discrete Laplacian operators of dimension 2^{m_x} and 2^{m_y} , respectively. Since T_x and T_y are supported on two disjoint set of qubits, they commute with each other, $[T_x, T_y] = 0$. Then,

$$\mathcal{U}(\delta t) = \exp \left[\frac{1}{h_x^2} (T_x \otimes I^{\otimes m_y}) \right] \exp \left[\frac{1}{h_y^2} (I^{\otimes m_x} \otimes T_y) \right]$$

Then this becomes two 1D heat equation Trotter steps inside one 2D heat equation Trotter steps. To avoid redundancy, we skip the circuit breakdown and proceed to generalize for dD heat equation.

check if CD count is 2^m and if there is some pattern in the angle parameters so that we can reduce it to m ,
check HHL paper

do a lit search to see if exponential speedup already exists

3.4.4 Example: d-dimensional Heat Equation

Let $\Omega \subset \mathbb{R}^d$ be a d -dimensional rectangular domain defined by:

$$\Omega = \{(x_1, x_2, \dots, x_d) \in \mathbb{R}^d : 0 < x_i < L_i \text{ for } i = 1, \dots, d\}$$

The boundary of the domain is denoted by $\partial\Omega$. Then the evolution of the temperature field $u(\mathbf{x}, t)$ is governed by the d -dimensional heat equation:

$$\frac{\partial u(\mathbf{x}, t)}{\partial t} = \alpha \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2} u(\mathbf{x}, t), \quad \mathbf{x} \in \Omega, \quad t > 0 \quad (61)$$

We impose homogeneous Dirichlet boundary conditions on all $2d$ faces of the hyper-prism:

$$u(\mathbf{x}, t) = 0, \quad \mathbf{x} \in \partial\Omega, \quad t \geq 0$$

For each dimension $i \in \{1, \dots, d\}$:

$$u(x_1, \dots, x_i = 0, \dots, x_d, t) = 0$$

$$u(x_1, \dots, x_i = L_i, \dots, x_d, t) = 0$$

The state of the system at $t = 0$ is given by the initial distribution function $f(\mathbf{x})$:

$$u(\mathbf{x}, 0) = f(x_1, x_2, \dots, x_d), \quad \mathbf{x} \in \Omega$$

This becomes d 1D heat equation Trotter steps inside one dD heat equation Trotter steps. The resource requirement (generalized from 1D) is given in the second row of table 1.

4 Implementation and experiment set-ups (MZ)

NOTE: in the qutip and bosonic qiskit, it is default $\hbar = 1$ but we previously used $\hbar = 2$, so some expressions in this section does not match up with the previous one since this is directly from my implementation notes. I will modified it.

Dimension	(qubit, qumode)	X gate	CD gate	Toffoli gate	Circuit Depth
1D	$(m + c, 1)$	$2n(m - 1)$	$3mn$	$3mn$	$\mathcal{O}(nm)$
dD	$(dm + c, 1)$	$d \times X(1d)$	$d \times CD(1d)$	$d \times \text{Toffoli}(1d)$	$d \times \text{depth}(1d)$

Table 1: Resource cost scaling as a function of Trotter steps n and qubit register size m , where $n = \mathcal{O}\left(\frac{t^{1+1/p}}{\epsilon_t^{1/p}}\right)$ is a function of runtime t and simulation accuracy ϵ_t , and qubit register size $m = \log_2 N$ is a function of grid points per dimension N . Here, $c \in \{1, 2, 3, 4\}$

The implemented hybrid map for runtime T is the CV-sandwich operator

$$K(T) = \langle \phi_r | e^{-iT \hat{x} \otimes A} | \psi_{r,r',\beta} \rangle, \quad (62)$$

where the post-selection state is a squeezed vacuum

$$|\phi_r\rangle = S(r) |0\rangle, \quad (63)$$

and the preparation state is a squeezed finite Fock superposition

$$|\psi_{r,r',\beta}\rangle = S(r') \sum_{n=0}^{N_{\max}-1} C_n |n\rangle. \quad (64)$$

Here $N_{\max}$ is the number of retained coefficients.

For a specific initial vector $\mathbf{u}(0)$, the post-selected DV output is $\mathbf{u}_{\text{cv}} = K(T)\mathbf{u}(0)$ and the reference is $\mathbf{u}_{\text{ref}} = e^{-AT}\mathbf{u}(0)$. We report the state fidelity

$$F = \left| \frac{\langle \mathbf{u}_{\text{cv}}, \mathbf{u}_{\text{ref}} \rangle}{\|\mathbf{u}_{\text{cv}}\| \|\mathbf{u}_{\text{ref}}\|} \right|^2, \quad (65)$$

and the post-selection probability

$$p_{\text{post}} = \|\mathbf{u}_{\text{DV,post}}\|_2^2, \quad (66)$$

where $\mathbf{u}_{\text{DV,post}}$ denotes the DV-register amplitudes immediately after projecting the CV mode onto $\langle \phi_r |$.

4.1 CV state preparation

We use a near-optimal kernel branch parameterized by $0 < \beta < 1$:

$$f_\beta(k) = \frac{e^{2\beta}}{2\pi e^{(1+ik)^\beta}}, \quad g_\beta(k) = \frac{f_\beta(k)}{1 - ik}. \quad (67)$$

Let $\sigma = e^r$ and $\sigma' = e^{r'}$ denote squeezing widths. Under the $\hbar = 1$ convention, the coefficient integral used by the numerical backends is

$$C_n \propto \int_{-\infty}^{\infty} H_n\left(\frac{k}{\sigma'}\right) g_\beta(k) e^{-\gamma k^2} dk, \quad \gamma = \frac{1}{2} (e^{-2r'} - e^{-2r}), \quad (68)$$

where $H_n(\cdot)$ is the physicists' Hermite polynomial. The integral is evaluated numerically using Gaussian quadrature. The state $|\psi_{\text{finite}}\rangle = \sum_{n=0}^{N_{\max}-1} C_n |n\rangle$ is prepared using both variational hybrid ansatz and Givens synthesis (Law-Eberly [12]). The variational ansatz is achieved through the SNAP-displacement ansatz in [16, 17],

$$U_{\text{SNAP+D}} = \prod_{\ell=1}^L D(\alpha_\ell) \text{SNAP}(\boldsymbol{\theta}_\ell), \quad (69)$$

optimized for

$$\max_{\{\alpha_\ell, \boldsymbol{\theta}_\ell\}} |\langle \psi_{\text{finite}} | U_{\text{SNAP+D}} | 0 \rangle|^2. \quad (70)$$

On the other hand, The Givens method prepares $|\psi_{\text{finite}}\rangle = \sum_{n=0}^{N-1} C_n |n\rangle$ from the vacuum $|0\rangle$ using $N - 1$ analytically determined rotation steps, without optimization. For Givens rotation, we define the two-level rotation acting on adjacent Fock levels $|n\rangle, |n+1\rangle$:

$$G(n, n+1; \theta, \varphi) = \begin{pmatrix} \cos \theta & -e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & \cos \theta \end{pmatrix} \oplus I_\perp, \quad (71)$$

where the 2×2 block acts on the $\{|n\rangle, |n+1\rangle\}$ subspace and $I_\perp$ is the identity on all other Fock levels. Any N -component target state can be factored as

$$|\psi_{\text{finite}}\rangle = G(N-2, N-1; \theta_{N-1}, \varphi_{N-1}) \cdots G(1, 2; \theta_2, \varphi_2) G(0, 1; \theta_1, \varphi_1) |0\rangle, \quad (72)$$

where the angles $\{(\theta_k, \varphi_k)\}_{k=1}^{N-1}$ are computed recursively from the coefficients C_n by a top-down elimination. On a coupled qubit–oscillator platform, a Law–Eberly synthesis alternates single-qubit rotations that set relative amplitudes/phases and a calibrated qubit–oscillator exchange interaction that transfers population between adjacent Fock levels. The exchange dynamics are generated in [18],

$$H_{\text{JC}} = \hbar\Omega(t)(\sigma_- a^\dagger + \sigma_+ a), \quad (73)$$

which couples $|e, n\rangle \leftrightarrow |g, n+1\rangle$ at the $\sqrt{n+1}$ -dependent rate $\Omega_n = \Omega_0\sqrt{n+1}$. Importantly, a single resonant JC “swap” pulse is not photon-number selective by itself; the effective selectivity required by the Givens factorization is achieved in practice by hardware calibration and, when available, photon-number-resolved qubit control in the dispersive regime. A representative experimental demonstration of deterministic resonator-state synthesis using such calibrated sequences is in [13].

4.2 Gate-level compilation of hybrid evolution

Given $A = \sum_m c_m P_m$, we use a first-order product formula over N_t time slices:

$$e^{-iT\hat{x}\otimes A} \approx \left(\prod_m e^{-i\Delta t c_m \hat{x}\otimes P_m} \right)^{N_t}, \quad \Delta t = \frac{T}{N_t}. \quad (74)$$

Each Pauli term is compiled by a basis change B_m (diagonalize P_m into Z -basis) followed by a parity mapping C_m onto a target qubit t :

$$e^{-i\Delta t c_m \hat{x}\otimes P_m} = B_m^\dagger C_m^\dagger e^{-i\Delta t c_m \hat{x}\otimes Z_t} C_m B_m. \quad (75)$$

The primitive hybrid interaction $e^{-i\lambda\hat{x}\otimes Z_t}$ is implemented as a conditional displacement with amplitude $\alpha = -i\lambda/\sqrt{2}$.

5 Experiments (MZ)

Need to sort the results out a little (comment out)

6 Conclusion

7 Future Works

8 Author Contributions

If needed

9 Acknowledgments

ERD acknowledges fruitful discussions with Kaustubh Chandramouli.

10 Competing interests

All authors declare no financial or non-financial competing interests.

11 Data and code availability

References

- [1] Laura Clinton, Toby Cubitt, Brian Flynn, Filippo Maria Gambetta, Joel Klassen, Ashley Montanaro, Stephen Piddock, Raul A. Santos, and Evan Sheridan. Towards near-term quantum simulation of materials. *Nature Communications*, 15(1), January 2024.
- [2] *Quantum Information and Computation*, 12(11 & 12), November 2012.
- [3] Dong An, Jin-Peng Liu, and Lin Lin. Linear combination of hamiltonian simulation for nonunitary dynamics with optimal state preparation cost. *Physical Review Letters*, 131(15):150603, 2023.
- [4] Dong An, Andrew M Childs, and Lin Lin. Quantum algorithm for linear non-unitary dynamics with near-optimal dependence on all parameters. *arXiv preprint arXiv:2312.03916*, 2023.
- [5] Jin-Peng Liu, Herman Oie Kolden, Hari K. Krovi, Nuno F. Loureiro, Konstantina Trivisa, and Andrew M. Childs. Efficient quantum algorithm for dissipative nonlinear differential equations. *Proceedings of the National Academy of Sciences*, 118(35):e2026805118, 2021.
- [6] Matthew Pocrnic, Peter D Johnson, Amara Katabarwa, and Nathan Wiebe. Constant-factor improvements in quantum algorithms for linear differential equations. *arXiv preprint arXiv:2506.20760*, 2025.
- [7] Luke Bell, Yan Wang, Kevin C Smith, Yuan Liu, Eugene Dumitrescu, and SM Girvin. Co-designing eigen- and singular-value transformation oracles: From algorithmic applications to hardware compilation. *arXiv preprint arXiv:2502.16029*, 2025.
- [8] Yuki Sato, Ruho Kondo, Ikko Hamamura, Tamiya Onodera, and Naoki Yamamoto. Hamiltonian simulation for hyperbolic partial differential equations by scalable quantum circuits. *Phys. Rev. Res.*, 6:033246, Sep 2024.
- [9] Hiran Alipanah, Feng Zhang, Yongxin Yao, Richard Thompson, Nam Nguyen, Junyu Liu, Peyman Givi, Brian J McDermott, and Juan José Mendoza-Arenas. Quantum dynamics simulation of the advection-diffusion equation. *arXiv preprint arXiv:2503.13729*, 2025.
- [10] Shi Jin and Nana Liu. Analog quantum simulation of partial differential equations. *Quantum Science and Technology*, 9(3):035047, June 2024.
- [11] Haiyong Wang and Lun Zhang. Convergence analysis of hermite approximations for analytic functions, 2025.
- [12] Chi K Law and Joseph H Eberly. Arbitrary control of a quantum electromagnetic field. *Physical review letters*, 76(7):1055, 1996.
- [13] Max Hofheinz, H Wang, Markus Ansmann, Radoslaw C Bialczak, Erik Lucero, Matthew Neeley, AD O’connell, Daniel Sank, J Wenner, John M Martinis, et al. Synthesizing arbitrary quantum states in a superconducting resonator. *Nature*, 459(7246):546–549, 2009.
- [14] Tanuj Khattar and Craig Gidney. Rise of conditionally clean ancillae for efficient quantum circuit constructions. *Quantum*, 9:1752, May 2025.
- [15] Fumiya Hanamura, Warit Asavanant, Hironari Nagayoshi, Atsushi Sakaguchi, Ryuhoh Ide, Kosuke Fukui, Peter van Loock, and Akira Furusawa. Implementing arbitrary multimode continuous-variable quantum gates with fixed non-gaussian states and adaptive linear optics. *Physical Review A*, 110(2), August 2024.
- [16] Reinier W Heeres, Brian Vlastakis, Eric Holland, Stefan Krastanov, Victor V Albert, Luigi Frunzio, Liang Jiang, and Robert J Schoelkopf. Cavity state manipulation using photon-number selective phase gates. *Physical review letters*, 115(13):137002, 2015.
- [17] Stefan Krastanov, Victor V. Albert, Chao Shen, Chang-Ling Zou, Reinier W. Heeres, Brian Vlastakis, Robert J. Schoelkopf, and Liang Jiang. Universal control of an oscillator with dispersive coupling to a qubit. *Phys. Rev. A*, 92:040303, Oct 2015.
- [18] Edwin T Jaynes and Frederick W Cummings. Comparison of quantum and semiclassical radiation theories with application to the beam maser. *Proceedings of the IEEE*, 51(1):89–109, 2005.
- [19] Dong An, Andrew M. Childs, Lin Lin, and Lexing Ying. Laplace transform based quantum eigenvalue transformation via linear combination of hamiltonian simulation, 2024.

A Appendix

A.1 Complementary (Homogeneous) Solution $\mathcal{T}e^{-\int_0^t A(s)ds}u_0$

Remark 1. With a quadrature rule, the goal is to find the approximation

$$\int_a^b g(x) \approx \sum_{j=0}^{m-1} w_j g(x_j). \quad (76)$$

For an integral over $[-1, 1]$ with Gaussian quadrature with m nodes, set node x_j to be the root of Legendre polynomial of degree m , $P_m(x)$ with weight $w_j = \frac{2}{(1-x_j^2)((P'_m(x_j))^2)}$. For a longer interval $[a, b]$, the composite rule is applied by dividing the interval $[a, b]$ into K segments, i.e., $[a + kh, a + (k+1)h]$ where $h = \frac{b-a}{K}$ is the step size. Gaussian quadrature is applied on each shorter interval and it is constructed from that on $[-1, 1]$ by a change of variable. That is, the weight $w_j \rightarrow \frac{h}{2}w_j$ and node $x_j \rightarrow \frac{h}{2}x_j + (a + kh) + \frac{1}{2}h$.

Consider the ODE solution over a truncated finite interval $[-K, K]$ using the composite Gaussian quadrature to discretize the interval.

The first step is to define

$$g(k) = \frac{1}{C_\beta(1 - ik)e^{(1+ik)^\beta}}, \quad U(T, k) = \mathcal{T}e^{-i \int_0^T (kL(s) + H(s))ds}. \quad (77)$$

Then,

$$\begin{aligned} \mathcal{T}e^{-\int_0^T A(s)ds} &= \int_{\mathbb{R}} g(k)U(T, k)dk && \text{(by LCHS)} \\ &\approx \int_{-K}^K g(k)U(T, k)dk && \text{(truncation)} \\ &= \sum_{m=-K/h_1}^{K/h_1-1} \int_{mh_1}^{(m+1)h_1} g(k)U(T, k)dk && \text{(divide to shorter intervals)} \\ &\approx \sum_{m=-K/h_1}^{K/h_1-1} \sum_{q=0}^{Q-1} c_{q,m}U(T, k_{q,m}) && \text{(Gaussian quadrature in each interval)} \end{aligned} \quad (78)$$

$$= \sum_{j=0}^{M-1} c_j U(T, k_j) \quad \text{(simplified notation)} \quad (79)$$

where h_1 is the step size in the composite Gaussian quadrature, Q is the number of nodes in each interval $[mh_1, (m+1)h_1]$, $k_{q,m}$ is a Gaussian node, $c_{q,m} = w_q g(k_{q,m})$, w_q is a Gaussian weight (which does not depend on the choice of m), and M is the overall number of the nodes (or unitaries).

Assumption 1. Assume K and h_1 are chosen so that K/h_1 is an integer.

To bound the truncation error by $\epsilon > 0$, it suffices to choose

$$K = \mathcal{O}\left((\log(1/\epsilon))^{1/\beta}\right)$$

To bound the quadrature error by $\epsilon > 0$, it suffices to choose

$$h_1 = \frac{1}{eT \max_t \|L(t)\|}, \quad Q = \left\lceil \frac{1}{\log 4} \log \left(\frac{8}{3C_\beta} \frac{K}{\epsilon} \right) \right\rceil = \mathcal{O}\left(\log \frac{1}{\epsilon}\right), \quad (80)$$

and the overall number of unitaries in the summation formula is

$$M = \frac{2KQ}{h_1} = \mathcal{O}\left(T \max_t \|L(t)\| \left(\log \frac{1}{\epsilon}\right)^{1+1/\beta}\right) \quad (81)$$

For the specific value of K , note that in Eq. (176) in [4], we have

$$\left\| \int_{\mathbb{R}} g(k)U(T, k)dk - \int_{-K}^K g(k)U(T, k)dk \right\| \leq \frac{2^{B+1}B!}{C_\beta(\cos(\beta\pi/2))^B} \frac{1}{K} e^{-\frac{1}{2}K^\beta \cos(\beta\pi/2)} \leq \epsilon$$

$$\ln \left(\frac{2^{B+1}B!}{C_\beta(\cos(\beta\pi/2))^B} \right) + \ln(1/\epsilon) \leq \ln(K) + \frac{1}{2}K^\beta \cos(\beta\pi/2) \quad (82)$$

$$2 \cdot \frac{\ln \left(\frac{2^{B+1}B!}{C_\beta(\cos(\beta\pi/2))^B} \right) + \ln(1/\epsilon) - \ln(K)}{\cos(\beta\pi/2)} \leq K^\beta \quad (83)$$

where $B = \lceil 1/\beta \rceil$

A.2 Inhomogeneous Term $\int_0^t \mathcal{T} e^{-\int_s^t A(s')ds'} b(s)ds$

The discretization of the inhomogeneous term is

$$\begin{aligned} & \int_0^T \mathcal{T} e^{-\int_s^T A(s')ds'} b(s)ds \\ &= \int_0^T \int_{\mathbb{R}} \frac{f(k)}{1-ik} U(T, s, k) b(s) dk ds \quad (\text{by LCHS Eq. (3)}) \\ &\approx \int_0^T \sum_{m_1=-K/h_1}^{K/h_1-1} \sum_{q_1=0}^{Q_1-1} c_{q_1, m_1} U(T, s, k_{q_1, m_1}) b(s) ds \quad (\text{by Eq. (78)}) \\ &\approx \sum_{m_2=0}^{T/h_2-1} \sum_{q_2=0}^{Q_2-1} \sum_{m_1=-K/h_1}^{K/h_1-1} \sum_{l_1=0}^{Q_1-1} c'_{q_2, m_2} c_{q_1, m_1} U(T, s_{q_2, m_2}, k_{q_1, m_1}) |b(s_{q_2, m_2})\rangle \quad (\text{composite Gaussian quadrature}) \quad (84) \end{aligned}$$

$$= \sum_{j'=0}^{M'-1} \sum_{j=0}^{M-1} c'_{j'} c_j U(T, s_{j'}, k_j) |b(s_{j'})\rangle \quad (\text{simplified notation}) \quad (85)$$

where $U(T, s, k) = \mathcal{T} e^{-i \int_s^T (kL(s') + H(s'))ds'}$, h_2 is the step size for the discretization of the time integral, Q_2 is the number of Gaussian quadrature nodes, s_{q_2, m_2} is a node, $c'_{q_2, m_2} = w'_{q_2} \|b(s_{q_2, m_2})\|$, the w'_{q_2} is a Gaussian weight. The $|b\rangle$ implies that it is the normalized vector $b/\|b\|$.

Let

- $\Lambda = \sup_{p \geq 0, t \in [0, T]} \|A^{(p)}\|^{1/(p+1)}$
- $\Xi = \sup_{p \geq 0, t \in [0, T]} \|b^{(p)}\|^{1/(p+1)}$, where (p) refers to the p^{th} -order time derivative
- $\|b\|_{L^1} = \int_0^T \|b(s)\| ds$

To bound the approximation error of Eq. (85) by $\epsilon > 0$, it suffices to choose

$$K = \mathcal{O} \left(\left(\log \left(1 + \frac{\|b\|_{L^1}}{\epsilon} \right) \right)^{1/\beta} \right), \quad h_1 = \frac{1}{eT \max_t \|L(t)\|}, \quad h_2 = \frac{1}{eK(\Lambda + \Xi)}, \quad (86)$$

$$Q_1 = \mathcal{O} \left(\log \left(1 + \frac{\|b\|_{L^1}}{\epsilon} \right) \right), \quad Q_2 = \mathcal{O} \left(\log \left(\frac{T(\Lambda + \Xi)}{\epsilon} \right) \right) \quad (87)$$

and

$$M' = \frac{TQ_2}{h_2} = \mathcal{O} \left(TK(\Lambda + \Xi) \log \left(\frac{T(\Lambda + \Xi)}{\epsilon} \right) \right) = \tilde{\mathcal{O}} \left(T(\Lambda + \Xi) \left(\log \left(1 + \frac{\|b\|_{L^1}}{\epsilon} \right) \right)^{1/\beta} \log(1/\epsilon) \right)$$

As indicated in Eq. (234) and (235) in [4], we have

$$\begin{aligned} \|c\|_1 \frac{eT(\Lambda + \Xi)}{4Q_2} &\leq \frac{\epsilon}{2} \\ \frac{1}{\ln(4)} \ln \left(\|c\|_1 \frac{2eT(\Lambda + \Xi)}{\epsilon} \right) &\leq Q_2, \end{aligned} \quad (88)$$

where $\|c\|_1 = \mathcal{O}(1)$. For computing Q_1 , just use the same formula in Eq. (80) but with the K in Eq. (86).

A.3 Summary

With LCHS, Eq. (2) becomes

$$u(T) \approx \sum_{j=0}^{M-1} c_j U(T, k_j) |u_0\rangle + \sum_{j'=0}^{M'-1} \sum_{j=0}^{M-1} c'_{j'} c_j U(T, s_{j'}, k_j) |b(s_{j'})\rangle. \quad (89)$$

where $U(T, s, k) = \mathcal{T}e^{-i \int_s^T (kL(s') + H(s')) ds'}$ and $U(T, k) = U(T, 0, k)$. So, we need oracles circuit to implement the Hamiltonian simulation operator $U(T, s, k)$ and oracles to encode the coefficients c and c' into amplitudes. Then, use LCU to obtain $u(T)$.

A.4 Analytical Values of the Expansion Coefficients C_n at limit values of β

The coefficients are defined by the projection of the signal $g(x)$ onto the Hermite basis:

$$C_n(\beta) = \mathcal{K}_n \int_{-\infty}^{\infty} \frac{H_n(x/\sigma') e^{-\gamma x^2}}{2\pi e^{-2\beta} e^{(1+ix)^\beta} (1-ix)} dx \quad (90)$$

where $\mathcal{K}_n = \sqrt{\frac{\sigma}{\sigma'}} \frac{1}{\sqrt{2^n n!}}$ and $\gamma = \sigma'^{-2} - \sigma^{-2}$. We denote the coefficient functions at the limit points as $C_n^{(0)}$ and $C_n^{(1)}$, corresponding to the limits $\beta = 0$ and $\beta = 1$, respectively.

We employ an explicit polynomial decomposition of the Hermite-Lorentzian kernel. By defining a residual quotient polynomial $Q_{n-1}(x) = [H_n(x/\sigma') - H_n(-i/\sigma')]/(x+i)$, we isolate the singular contribution of the complex pole at $x = -i$.

At the limit $\beta = 0$, the denominator simplifies to a standard complex Lorentzian. The exact closed-form expression is:

$$C_n^{(0)} = \mathcal{K}_n \left[\frac{1}{2} H_n \left(-\frac{i}{\sigma'} \right) e^{\gamma} \operatorname{erfc}(\sqrt{\gamma}) + \frac{i}{2\pi} \int_{-\infty}^{\infty} Q_{n-1}(x) e^{-\gamma x^2} dx \right] \quad (91)$$

For $\beta = 1$, the kernel incorporates a phase shift e^{-ix} and a scaling factor e^1 . Completing the square in the exponent for the shifted Gaussian $(-\gamma x^2 - ix)$ leads to:

$$C_n^{(1)} = \mathcal{K}_n \left[\frac{1}{2} H_n \left(-\frac{i}{\sigma'} \right) e^{\gamma} \operatorname{erfc} \left(\sqrt{\gamma} - \frac{1}{2\sqrt{\gamma}} \right) + \frac{ie}{2\pi} \int_{-\infty}^{\infty} Q_{n-1}(x) e^{-\gamma x^2 - ix} dx \right] \quad (92)$$

The transition from $\beta = 0$ to $\beta = 1$ effectively shifts the argument of the error function by $\Delta = (2\sqrt{\gamma})^{-1}$.

A.4.1 Numerical Validation

The analytical results were compared against direct numerical quadrature using a double-exponential integration scheme with a working precision of 30 decimal places. For the parameter set $\sigma' = 1.2$ and $\sigma = 2.0$ ($\gamma = 4/9$), the agreement is presented in the tables below.

Table 2: Comparison of Numerical and Analytical Coefficients for the Lower Bound ($\beta = 0, \gamma = 4/9$)

n	Numerical Result	Analytical (Explicit)
0	0.3481065885	0.3481065885
1	0.2335427879 <i>i</i>	0.2335427879 <i>i</i>
2	-0.0515295394	-0.0515295394
3	-0.0077856810 <i>i</i>	-0.0077856810 <i>i</i>

Table 3: Comparison of Numerical and Analytical Coefficients for the Upper Bound ($\beta = 1, \gamma = 4/9$)

n	Numerical Result	Analytical (Explicit)
0	1.1011786413	1.1011786413
1	-0.3006294949 <i>i</i>	-0.3006294949 <i>i</i>
2	-0.0943736080	-0.0943736080
3	-0.1769037943 <i>i</i>	-0.1769037943 <i>i</i>

As demonstrated in Tables 2 and 3, the explicit decomposition method maintains a relative error below 10^{-25} .

B Fourier-Laplace Transform

Ref. [19] proposed an LCHS-based Laplace transform QEVT. The theorem is as follows:

Let

$$A = L + iH, \quad \text{where} \quad L = \frac{A + A^\dagger}{2} \succeq 0 \quad \text{and} \quad H = \frac{A - A^\dagger}{2i}$$

are both Hermitian matrices.

For a function

$$h(z) = \int_0^\infty g(t) e^{-zt} dt$$

that can be expressed by a Laplace transformation with $g(t)$ being its inverse Laplace transform, we have

$$h(A) = \int_0^\infty \int_{\mathbb{R}} \frac{f(k)g(t)}{1 - ik} e^{-it(kL+H)} dk dt. \quad (5)$$

Here

$$f(k) = \frac{1}{2\pi e^{-2\beta} e^{(1+iz)\beta}}, \quad \beta \in (0, 1).$$

We propose the CVDV LCHS based Laplace transform theorem as follows:

Theorem 3.

$$|u(t)\rangle = h(A) |u(0)\rangle = \langle \Phi_{\text{osc}_1} | \langle \Phi_{\text{osc}_2} | e^{-i(\hat{x}_1 \otimes \hat{x}_2 \otimes L + \hat{x}_1 \otimes \mathbf{I} \otimes H)} | \Psi_{\text{osc}_2} \rangle | \Psi_{\text{osc}_1} \rangle \quad (93)$$

$$= (\langle \phi |_{\text{osc}_1} \otimes \mathbf{I}_{\text{osc}_2} \otimes \mathbf{I}_q) (\mathbf{I}_{\text{osc}_1} \otimes \langle \phi |_{\text{osc}_2} \otimes \mathbf{I}_q) e^{-it(\hat{x} \otimes L + \mathbf{I}_{\text{osc}} \otimes H)} (|\psi\rangle_{\text{osc}_1} \otimes |\psi\rangle_{\text{osc}_2} \otimes |u_0\rangle_q) \quad (94)$$

$$= \left[\int_{\mathbb{R}} \int_{\mathbb{R}} dx_1 dx_2 \phi_1^*(x_1) \psi_1(x_1) \phi_2^*(x_2) \psi_2(x_2) e^{-it(xL+H)} \right] |u_0\rangle_q. \quad (95)$$